Monday Coffee

Business Expansion Analysis

Please read all instructions carefully before you begin.

Project Overview

Monday Coffee is a fictional coffee brand that has been selling its products online across multiple Indian cities since January 2023. The company is now looking to expand into physical store locations and wants to make data-driven decisions about where to open.

In this capstone project, you will act as a Data Analyst for Monday Coffee. Using SQL, you will analyse their sales, customer, and city data to identify the top three Indian cities best suited for new physical coffee shop locations.

Dataset Files

You have been provided with four CSV files. Import them into PostgreSQL (or your preferred SQL environment) before you begin.

File	Description
`city.csv`	City population, rent, and rank data
`products.csv`	Coffee product names and prices
`customers.csv`	Customer records and city links
`sales.csv`	All sales transactions

Instructions & Submission Guidelines

General Rules

- Use PostgreSQL for all queries.
- All queries must run without errors. Test each one before submission.
- Use meaningful aliases for tables and columns to improve readability.
- Add at least one comment (--) per query to explain your logic.
- Do not hard-code values that can be derived from the data (e.g., avoid typing city names directly unless the question requires it).

Core Expectation

- Understand the problem and define clear objectives

SQL Questions

Answer all 10 questions below.

Question 1: **Coffee Consumer Estimate**

Assuming 25% of each city's population drinks coffee, calculate the estimated number of coffee consumers (in millions) per city. Order results from highest to lowest.

Question 2: **Total Revenue - Q4 2023**

What is the total revenue generated from coffee sales across all cities during the last quarter of 2023 (October-December)? Show results per city, ordered by revenue descending.

Question 3: **Sales Volume by Product**

How many units of each coffee product have been sold in total? Rank products from best-selling to least-selling.

Question 4: **Average Sales per Customer by City**

What is the average total sales amount per unique customer in each city? Include total revenue and customer count alongside the average. Order by total revenue descending.

Question 5: **Current Customers vs. Estimated Coffee Consumers**

For each city, show both the estimated coffee-drinking population (25% of city population, in millions) and the actual number of unique customers from the sales data. Use a CTE.

Question 6: **Top 3 Products per City**

What are the top 3 best-selling coffee products in each city, based on number of orders? Use a window function to rank products within each city.

Question 7: **Unique Customers per City**

How many unique customers in each city have made at least one coffee purchase? Order by customer count descending.

Question 8: **Average Sale vs. Average Rent per Customer**

For each city, compare the average sale amount per customer against the average rent cost per customer (estimated_rent divided by number of customers). This helps evaluate cost efficiency.

Question 9: **Month-on-Month Sales Growth**

Calculate the month-on-month percentage change in total sales for each city. Use a window function (LAG) to compare each month's sales to the previous month. Show only rows where a prior month exists.

Question 10: **Market Potential Summary**

Produce a full market potential table for each city, showing: total revenue, estimated rent, total customers, estimated coffee consumers (millions), average sale per customer, and average rent per customer. Order by total revenue descending.

Bonus Task: Design Your Own Questions

Now that you've explored the Monday Coffee dataset, it's your turn to think like an analyst. Come up with three original business questions that can be answered using SQL on this dataset. For each question:

1. Write the business question clearly (what insight are you trying to surface?).
2. Write the SQL query that answers it.
3. Write a one-sentence interpretation of what the result tells Monday Coffee.

Final task: Recommendation

Based on your SQL analysis above, write your business recommendation. Identify the three cities you would recommend for Monday Coffee's first physical store locations and justify each choice using specific metrics from your queries (e.g., revenue, customer count, rent efficiency, consumer potential).

Deliverables

- Technical Report (README)

  Each PERSON must write a detailed technical report covering:
    - Problem statement
    - Data description
    - Methodology (questions and sql answers)
    - Results (screenshot of their output)
    - SQL concepts used
    - Key insights and recommendations
    - Limitations and future work
- GitHub Repository link
    - Each person must create a public GitHub repository containing:
    - Clean, well-documented .sql script containing their solutions
    - The Technical report README.md explaining the project
    - Upload your datasets to the repository